

10.5 Distance of a Point From a Line

The distance of a point from a line is the length of the perpendicular drawn from the point to the line. Let $L : Ax + By + C = 0$ be a line, whose distance from the point $P(x_1, y_1)$ is d . Draw a perpendicular PM from the point P to the line L (Fig10.19). If

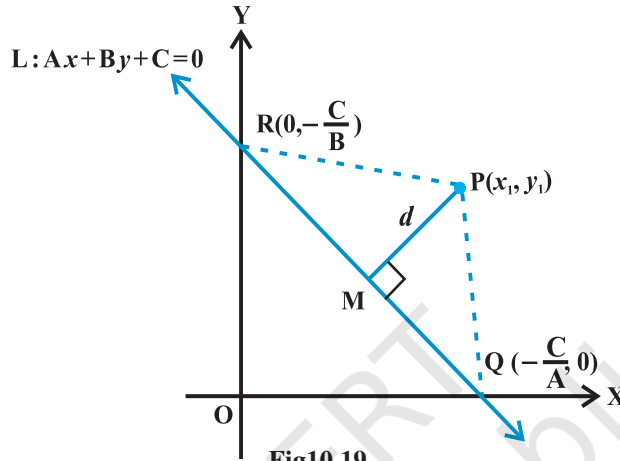


Fig10.19

the line meets the x - and y -axes at the points Q and R , respectively. Then, coordinates of the points are $Q\left(-\frac{C}{A}, 0\right)$ and $R\left(0, -\frac{C}{B}\right)$. Thus, the area of the triangle PQR is given by

$$\text{area } (\Delta PQR) = \frac{1}{2} PM \cdot QR, \text{ which gives } PM = \frac{2 \text{ area } (\Delta PQR)}{QR} \quad \dots (1)$$

$$\begin{aligned} \text{Also, area } (\Delta PQR) &= \frac{1}{2} \left| x_1 \left(0 + \frac{C}{B} \right) + \left(-\frac{C}{A} \right) \left(-\frac{C}{B} - y_1 \right) + 0(y_1 - 0) \right| \\ &= \frac{1}{2} \left| x_1 \frac{C}{B} + y_1 \frac{C}{A} + \frac{C^2}{AB} \right| \end{aligned}$$

$$\text{or } 2 \text{ area } (\Delta PQR) = \left| \frac{C}{AB} \right| \cdot |Ax_1 + By_1 + C|, \text{ and}$$

$$QR = \sqrt{\left(0 + \frac{C}{A} \right)^2 + \left(\frac{C}{B} - 0 \right)^2} = \left| \frac{C}{AB} \right| \sqrt{A^2 + B^2}$$

Substituting the values of area (ΔPQR) and QR in (1), we get

$$PM = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

or

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}.$$

Thus, the perpendicular distance (d) of a line $Ax + By + C = 0$ from a point (x_1, y_1) is given by

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}.$$

10.5.1 Distance between two

parallel lines We know that slopes of two parallel lines are equal.

Therefore, two parallel lines can be taken in the form

$$y = mx + c_1 \quad \dots (1)$$

and $y = mx + c_2 \quad \dots (2)$

Line (1) will intersect x -axis at the point

$A\left(-\frac{c_1}{m}, 0\right)$ as shown in Fig10.20.

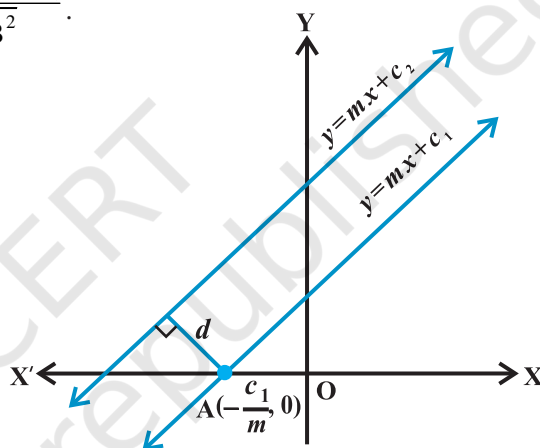


Fig10.20

Distance between two lines is equal to the length of the perpendicular from point A to line (2). Therefore, distance between the lines (1) and (2) is

$$\frac{\left| \left(-m\right)\left(-\frac{c_1}{m}\right) + (-c_2) \right|}{\sqrt{1+m^2}} \quad \text{or} \quad d = \frac{|c_1 - c_2|}{\sqrt{1+m^2}}.$$

Thus, the distance d between two parallel lines $y = mx + c_1$ and $y = mx + c_2$ is given by

$$d = \frac{|c_1 - c_2|}{\sqrt{1+m^2}}.$$

If lines are given in general form, i.e., $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$,

then above formula will take the form $d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$

Students can derive it themselves.

Example 18 Find the distance of the point $(3, -5)$ from the line $3x - 4y - 26 = 0$.

Solution Given line is $3x - 4y - 26 = 0$... (1)

Comparing (1) with general equation of line $Ax + By + C = 0$, we get

$$A = 3, B = -4 \text{ and } C = -26.$$

Given point is $(x_1, y_1) = (3, -5)$. The distance of the given point from given line is

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}} = \frac{|3 \cdot 3 + (-4)(-5) - 26|}{\sqrt{3^2 + (-4)^2}} = \frac{3}{5}.$$

Example 19 Find the distance between the parallel lines $3x - 4y + 7 = 0$ and

$$3x - 4y + 5 = 0$$

Solution Here $A = 3, B = -4, C_1 = 7$ and $C_2 = 5$. Therefore, the required distance is

$$d = \frac{|7 - 5|}{\sqrt{3^2 + (-4)^2}} = \frac{2}{5}.$$

EXERCISE 10.3

- Reduce the following equations into slope - intercept form and find their slopes and the y - intercepts.
 - $x + 7y = 0$,
 - $6x + 3y - 5 = 0$,
 - $y = 0$.
- Reduce the following equations into intercept form and find their intercepts on the axes.
 - $3x + 2y - 12 = 0$,
 - $4x - 3y = 6$,
 - $3y + 2 = 0$.
- Reduce the following equations into normal form. Find their perpendicular distances from the origin and angle between perpendicular and the positive x-axis.
 - $x - \sqrt{3}y + 8 = 0$,
 - $y - 2 = 0$,
 - $x - y = 4$.
- Find the distance of the point $(-1, 1)$ from the line $12(x + 6) = 5(y - 2)$.
- Find the points on the x-axis, whose distances from the line $\frac{x}{3} + \frac{y}{4} = 1$ are 4 units.
- Find the distance between parallel lines
 - $15x + 8y - 34 = 0$ and $15x + 8y + 31 = 0$
 - $l(x + y) + p = 0$ and $l(x + y) - r = 0$.